Annex B - Tan I-Je

Quality of Life & Social Indicators

1. Focus Area and Analytical Objective

Focus Area

Quality of Life & Social Indicators

Guiding Question

How liveable is Singapore beyond income?

While Singapore consistently ranks highly on economic indicators such as GDP per capita and income levels, these metrics alone do not fully capture residents' lived experiences. This analysis evaluates Singapore's quality of life using a **multi-dimensional framework** encompassing **health, education, safety, and subjective well-being**, benchmarked against selected regional and global peers.

The objective is to assess whether Singapore's strong economic and institutional performance translates into broader social outcomes that support long-term liveability and sustainability.

2. Data Sources and Indicators

Multiple authoritative global datasets were integrated to ensure reliability, comparability, and relevance:

Dimension	Indicator	Source	URL
Health	Life Expectancy at Birth	World Bank	https://data.worldbank.org/indicator/SP.DYN.LE00.IN
Health	Healthcare Expenditure per Capita (PPP)	World Bank	https://data.worldbank.org/indicator/SH.XPD.CHEX.PP.CD
Education	Educational Attainment (Average Years)	World Bank	https://data.worldbank.org/indicator/SE.SEC.CUAT.UP.ZS
Safety	Homicide Rate (per 100,000 population)	Our World in Data (UNODC)	https://data.worldbank.org/indicator/VC.IHR.PSRC.P5
Well-being	Life Satisfaction (Cantril Ladder)	Our World in Data	https://ourworldindata.org/grapher/happiness-cantril-ladder
Well-being	Composite Quality of Life Index	Derived	Constructed in Power BI

3. Data Preparation and Transformation

3.1 Data Cleansing

All datasets were imported into Power BI using Power Query and subjected to systematic cleaning procedures:

- Rows with missing or incomplete observations were removed to prevent distortions in cross-country and time-series analysis.
- Duplicate records and non-country aggregates were excluded.
- Column names and data types were standardised to ensure consistency across datasets.
- Temporal alignment was applied to ensure indicators were comparable across overlapping years.

These steps ensured that all retained observations were complete, comparable, and analytically valid.

3.2 Indicator Selection and Refinement

During exploratory analysis, multiple education indicators were reviewed. Educational attainment (average years of education) was retained as the primary education outcome measure because it:

- Reflects **long-term human capital accumulation**,
- Is less sensitive to short-term cohort effects,
- Enables meaningful comparison across developed economies.

Healthcare expenditure was treated as an **input indicator**, while life expectancy was retained as the primary **health outcome indicator**, consistent with best practices in social indicator analysis.

3.3 Benchmark Countries

To provide meaningful international context, Singapore was benchmarked against a selected group of comparable developed economies:

- **Singapore**
- **Japan**
- **Australia**
- **South Korea**
- **World (aggregate reference)**

4. Data Transformation and Normalisation

4.1 Base Metrics

For each indicator, average values were computed to summarise long-run performance:

- Average life expectancy
- Average healthcare spending per capita
- Average education attainment
- Average homicide rate
- Average life satisfaction

4.2 Normalisation of Indicators

As indicators are measured in different units (years, monetary values, rates, survey scores), **min–max normalisation** was applied to convert all metrics to a common **0–100 scale**:

$$\text{Normalised Score} = \frac{x - \min(x)}{\max(x) - \min(x)} \times 100$$

Safety Indicator Adjustment

Homicide rate is a negative indicator where lower values imply better outcomes. After normalisation, the safety score was inverted so that higher values consistently represent better performance:

$$\text{Safety Score} = (1 - \text{Normalised Homicide Rate}) \times 100$$

This ensures interpretive consistency across all quality-of-life dimensions.

5. Construction of the Composite Quality of Life Index

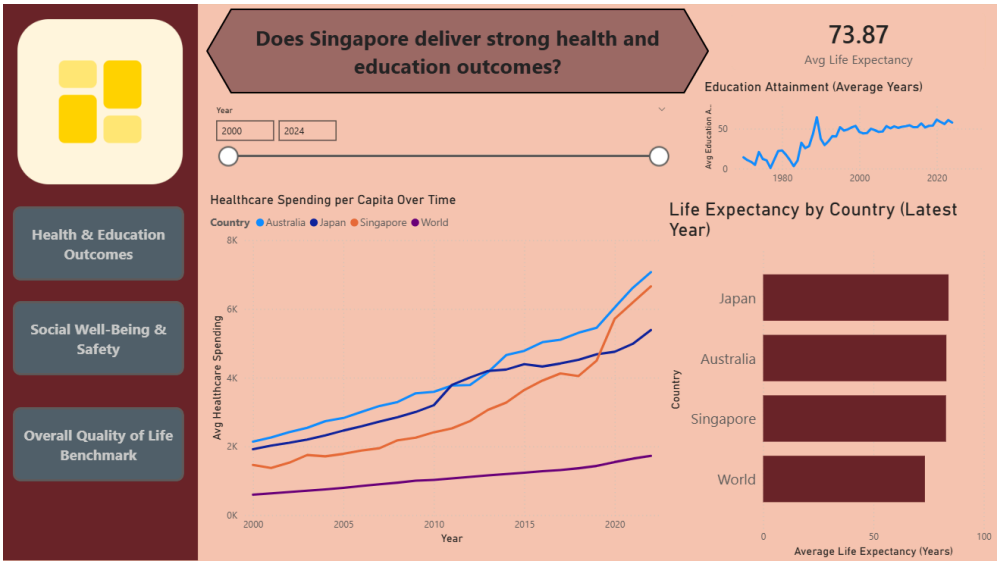
To provide an overall benchmark of liveability, a composite **Quality of Life (QoL) Index** was constructed by averaging the four normalised dimensions:

Health, education, safety, life satisfaction

Each dimension was assigned **equal weight**, reflecting the assumption that all aspects contribute equally to overall quality of life. This transparent construction allows users to interpret both individual components and the overall index without obscuring underlying trade-offs.

6. Data Analysis and Visualisation

6.1 Health and Education Outcomes



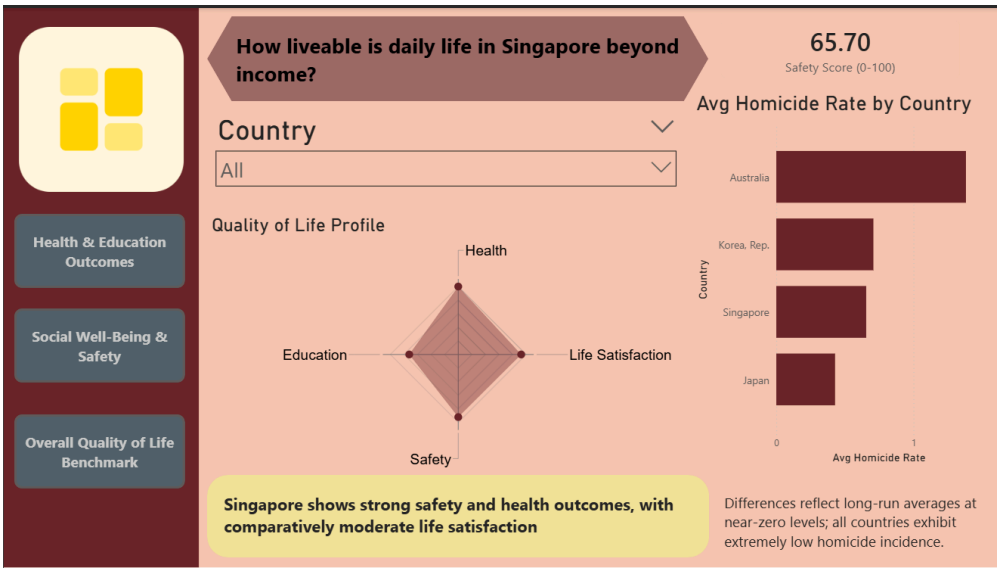
This section evaluates whether Singapore delivers strong foundational social outcomes.

- **Healthcare spending per capita over time** shows a steady upward trend, reflecting sustained investment in healthcare systems.

- **Life expectancy benchmarking (latest year)** indicates that Singapore performs strongly relative to regional peers and global averages.
- **Education attainment over time** demonstrates consistent improvement in long-run human capital development.

Together, these visuals suggest that Singapore's institutional investments translate into strong health and education outcomes.

6.2 Social Well-being and Safety

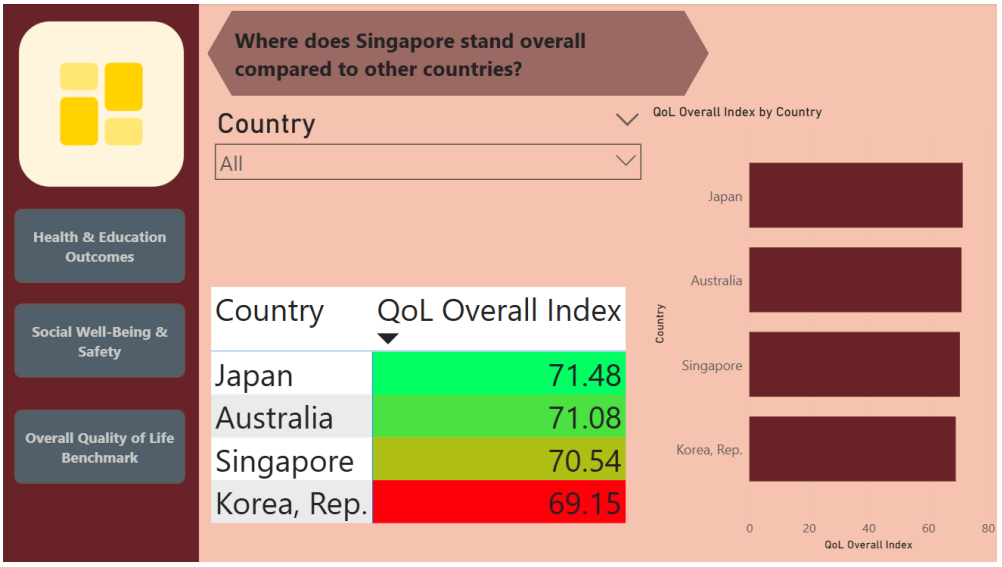


A radar chart is used to compare the four normalised QoL dimensions for Singapore:

- Safety emerges as a key strength, reflecting extremely low homicide rates.
- Health outcomes remain strong.
- Education outcomes are robust.
- Life satisfaction moderates overall performance, indicating that high income and strong infrastructure do not fully translate into higher subjective well-being.

This visual highlights the multi-dimensional nature of liveability and the trade-offs between objective outcomes and subjective experiences.

6.3 Overall Quality of Life Benchmarking



The composite QoL Index is used to benchmark Singapore against selected peers:

- A ranking table and bar chart compare overall QoL scores across countries.
- Singapore performs competitively but does not rank uniformly highest across all dimensions.
- Relatively lower life satisfaction moderates Singapore's overall ranking despite strong health, education, and safety outcomes.

This reinforces the importance of considering multiple dimensions when evaluating national liveability.

7. Key Insights

- Singapore demonstrates **strong performance in health and safety**, reflecting effective public systems and governance.
- Education outcomes remain robust, supporting long-term human capital development.
- Subjective well-being moderates Singapore's overall quality-of-life ranking, suggesting that economic success and infrastructure alone do not guarantee higher life satisfaction.
- Benchmarking against regional and global peers provides valuable context for evaluating long-term sustainability and liveability.

8. Conclusion

Singapore remains highly liveable beyond income, with particularly strong health, education, and safety outcomes. However, subjective well-being plays a critical moderating role in overall quality of life. These findings highlight important trade-offs for policymakers seeking to enhance long-term liveability, suggesting that future efforts should balance economic performance with social and psychological well-being.